

Dataset Card — Ultron Tool-Calling Fine-Tuning Dataset

Overview

Name	ultron-pentest-toolcall-v1
Purpose	Teach a base LLM to select and correctly format calls to Ultron's pentesting tool suite
Size (this run)	40,580 tokens processed across 30 training steps (~1.31 epochs)
Format	Instruction / tool-call pairs (chat-style, with <code>tool_calls</code> structured output)
Split	Training only — no validation/test split was carved out for this run (see Limitations)

Composition

Fill in the exact numbers for your submission — the structure below is what judges will expect to see documented.

- **Number of examples:** [fill in]
- **Average example length:** [fill in tokens]
- **Tool categories represented:**
 - Reconnaissance / scanning invocations (e.g., host/port/service discovery)
 - Vulnerability lookup and enrichment calls (e.g., CVE/CWE reference queries)
 - Reporting and documentation calls (e.g., generate finding, generate summary)
 - Session/state management calls (e.g., log step, request human confirmation)
- **Multi-turn vs. single-turn:** [fill in ratio]
- **Languages:** English

Provenance

- **Source:** Custom-authored for this project; [state whether hand-written, templated/synthetically generated, derived from public CTF writeups, or a mix]
- **Synthetic data generation** (if used): [name the generator model/process and note any filtering/review step applied]
- **Real target data:** [confirm whether any real scan output, real hostnames/IPs, or real client data appears in the dataset — if so, describe redaction/anonymization steps taken]

Labeling / Construction Process

[Describe how tool-call examples were constructed — e.g., manually written by the team, generated then reviewed, or derived from tool documentation]. Each example pairs a natural-language instruction with the exact structured tool call (tool name + arguments) the model should emit.

Licensing

- **This dataset's license:** [fill in — e.g., project-internal / CC-BY / other]
- **Any third-party content included:** [list sources and confirm redistribution rights, or confirm none was used]

Known Limitations

- Small scale (single 30-step run, ~40K tokens) — not yet large enough to guarantee generalization across all tool categories.
- No held-out validation/test split exists yet; see `EVALUATION_REPORT.md` for the planned fix.
- Tool-call examples reflect the current fixed tool schema (`TOOL_SCHEMA_REFERENCE.md`); adding or changing a tool requires new training examples for that tool before the model can be expected to call it correctly.
- Has not been checked for duplicate or near-duplicate examples across the set.

Ethical Considerations

The dataset intentionally avoids embedding real exploit payloads, working malware, or step-by-step attack instructions for specific CVEs — tool-call examples specify *which tool to invoke and with what parameters*, not the underlying attack technique. Any example derived from a real engagement must have client-identifying details (hostnames, IPs, credentials) removed before

inclusion.